

Eduardo Venegas

Data Engineer

EDUCATION

MSc. Data Science and Artificial Intelligence Universität des Saarlandes 10/2024 – 09/2026 Saarbrücken, Germany

- Coursing degree. *Statistics | Deep Learning | NLP | Machine Translation | Image Processing | Computer Vision.*
- Master Thesis:** BMW Group & DFKI

BSc. Computer Science Tecnológico de Monterrey GDL

08/2019 – 12/2023 Guadalajara, Mexico

- Grade: 94.6 / 100. Five times best software engineering GPA. **60% tuition** scholarship due to academic merit.
- ML | Time series | Algorithms & Optimization | Cloud Computing & Distributed | Automation & PLC | Vehicle User Interfaces.*

Computer Science Exchange Student Universität Ulm

10/2022 – 09/2023 Ulm, Germany

- Full scholarship** recipient from DAAD. *Computer Vision II | Autonomous Driving and Assistance Systems | Statistics*

Technologist in Automation and Robotics CETI Colomos

08/2015 – 06/2019 Guadalajara, Mexico

- Thesis Project:** Autonomous Tennis Machine. *Electronics | Robotics | Industrial Automation | Electricity | PLCs*

SKILLS

Data

RAG LangChain HuggingFace MCP R Matlab SQL TensorFlow Knowledge Graphs

Software

JavaScript TypeScript Python C C++ React Angular Flask Node Ionic/Tauri

Systems

Linux Docker Kubernetes AWS CI/CD Grafana MQTT PLC Arduino

WORK EXPERIENCE

Data Scientist Masterarbeit BMW Group & DFKI & Saarland University

03/2026 – 09/2026 Munich, Germany

- Topic:** *Evaluating Multi-Turn Tool-Using Agentic RAG Systems: A Synthetic Benchmark Generation Approach.*
- Architected a modular evaluation framework for **tool-using Agentic RAG** systems to enable fine-grained failure localization.
- Built a **multi-turn synthetic benchmark** that combines deterministic **task planning** with diverse user **personas**.
- Engineered an **unsupervised retrieval engine** to audit a **RAG system** over **4M unannotated** corporate records.

Data Scientist Werkstudent August-Wilhelm Scheer Institute

10/2024 – 02/2026 Saarbrücken, Germany

- Developed a **KG-RAG** over unstructured technical documentation and a **Planning Agent**, using **LangChain** and **HuggingFace**.
- Implemented an **Autoencoder** model for **Anomaly Detection** via **Federated Learning** across edge devices.

Cloud Software Engineer Oracle

11/2023 – 10/2024 Guadalajara, Mexico

- Worked on the **Oracle Cloud Infrastructure** in the team of **Big Data Service** in **Oracle Analytics**.
- Developed tools for managing and deploying **big data clusters**. Implemented **pipelines** for automation and testing.

Data Analyst Intern BMW Group

03/2023 – 09/2023 Munich, Germany

- Developed a high-accuracy algorithm for **detecting external battery replacements**.
- Found **root cause** to **missing database values**, created a new processing pipeline and **increasing service demand by 62%**.
- Designed a **Streamlit dashboard** for adjusting prediction model parameters and obtaining a **real-time output (<8 seconds)**.
- Configured **cloud-based environments** for working with **PySpark**.

Site Reliability Engineer Xira

01/2022 – 04/2022 Mexico City, Mexico

- Managed** over 10 servers, **monitored** services and **enhanced** engineering processes.
- Built a **platform** to manage client **APIs**, integrating payments and subscription systems using **NextJS**, **Stripe** and **Docker**.

Production Engineer Fellow Major League Hacking & Meta

06/2021 – 09/2021 New York, USA

- Selected among 120 out of 20,000 applicants. Implemented monitoring with **Prometheus** and **Grafana**.
- Automated **CI/CD** workflow for efficient deployment to a production server using Docker in an **AWS EC2** instance

Full Stack & Data Engineer Jiit Technologies

10/2020 – 06/2021 Guadalajara, Mexico

- Designed a custom sales data analysis pipeline API using **Flask**, **SciPy** and **Prophet**.
- Built an **ERP** platform from scratch to manage inventory, human resources, shipments, income and expenses. **Angular**.

PROJECTS

Daten-Mapping-Framework für die AAS

AAS

Industry 4.0

LLM

DOI: 10.17560/atp.v68i5.2836

- Developed a mapping **framework** from schemas to **Asset Administration Shell** to achieve interoperability in **Industry 4.0**.
- Designed a dual alignment strategy separating standard **relational** tables from **hierarchical** assets.

See Your World

Tensorflow

Google Cloud

Flask

Github

- Crafted a hackathon **award-winning** app for the **visually impaired**, supporting **11 languages**.
- Engineered **Image Captioning** DL model with **TensorFlow**, deployed using **Flask**, **Docker**, and **Google Cloud VM**.

DevUp

Docker

AWS

Monitoring

Github

- Secured **victory** in MLH Production Engineering **Hackathon**. Platform to help developers find teammates for side projects.
- Used **Ionic React**, **Docker**, **AWS EC2**, **NGINX**, **Flask**, **PostgreSQL**, **CI/CD**, **cAdvisor**, **Grafana** and **Prometheus**.

CERTIFICATIONS

- Mathematics for Data Science Specialization
- Data Science Professional Certificate
- Python Data Science Professional Certificate
- Deep Learning Professional Certificate

Higher School of Economics - 2021

HarvardX - 2020

IBMX - 2020

IBMX - 2020

LANGUAGES

Spanish Native

English C1

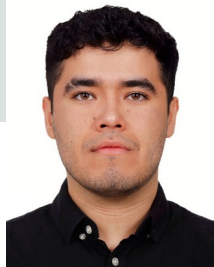
German C1

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